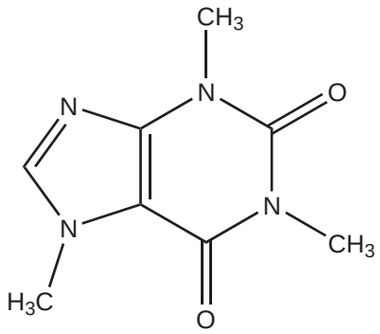
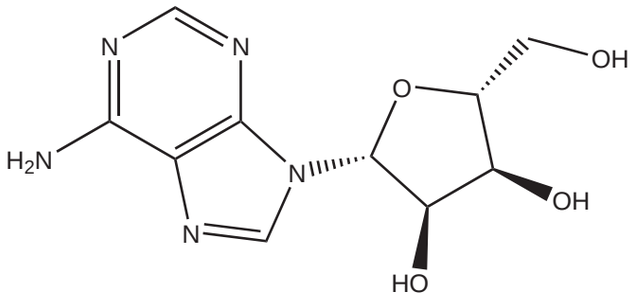
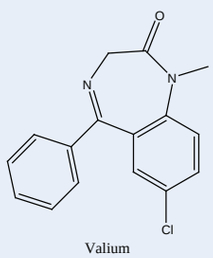
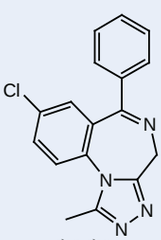
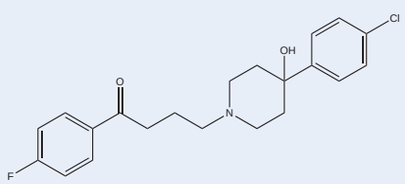




ADENOSINE

Therapeutic action of Different classes of Drugs:

The developments in the field of biology allowed us to understand various biological process and their mechanism in detail. This enabled to develop new safer efficient drugs. For example, to treat acidity, we have been using weak bases such as aluminium and magnesium hydroxides. But these can make the stomach alkaline and trigger the production of much acid. Moreover, This treatment only relives the symptoms and does not control the cause. Detailed studies reveal that histamines stimulate the secretion of HCl by activating the receptor in the stomach wall. This findings lead to the design of new drugs such as cimetidine, ranitidine etc.. which binds the receptor and inactivate them. These drugs are structurally similar to histamine.In this section, we shall discuss the therapeutic action of a few important classes of drugs.

Class of Drugs	Mode of action	Chemical structure of some important structures
1) Tranquilizers They are neurologically active drugs. **i) Major tranquilizers:** Haloperidol, clozapine **ii) Minor tranquilizers:** Diazepam (Valium), alprazolam	Acts on the central nervous system by blocking the neurotransmitter dopamine in the brain **Uses** Treatment of stress, anxiety, depression, sleep disorders and severe mental diseases like schizophrenia	 Valium  alprazolam  Haloperidol

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